CE 400/500

Special Topics: AI for Chemical Engineers

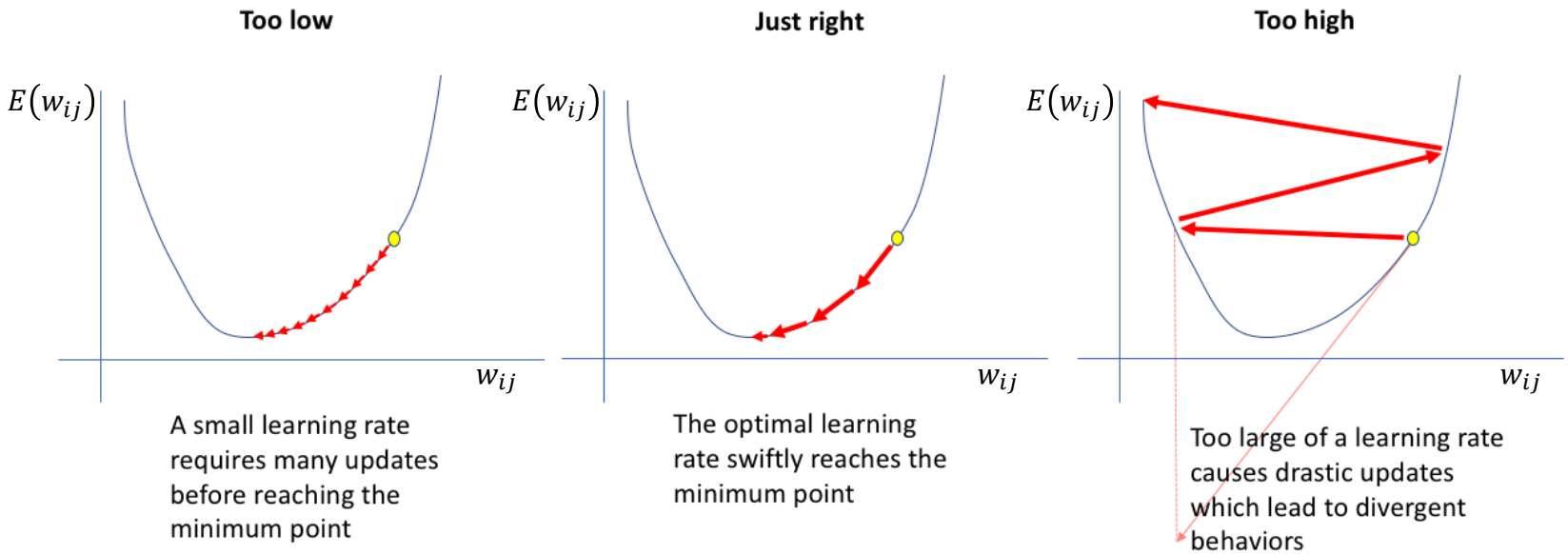
Lecture 11: Introduction to Neural Networks, Hands-on Practice to Predict Hydrogen Storage in MOFs with Multilayer Perceptron

University at Buffalo
The State University of New York

Learning Rate

$$w_{ij}^{(t+1)} = w_{ij}^{(t)} - \alpha \frac{dE}{dw_{ij}^{(t)}}$$

Learning rate (hyperparameter)



Adapted from the image by Jeremy Jordan

Let's try this interactive demo to learn the effect of learning rate.

- Pick "cos" dataset, try different learning rates, keep the same layer.

Best Practices for Neural Network Training

Vanishing Gradients

The gradients for the lower neural network layers (those closer to the input layer) can become very small. In deep networks (networks with more than one hidden layer), computing these gradients can involve taking the product of many small terms. When the gradient values approach 0 for the lower layers, the gradients are said to "**vanish**". Layers with vanishing gradients train very slowly, or not at all.

Solution: The ReLU activation function can help prevent vanishing gradients.

$$w_{ij}^{(t+1)} = w_{ij}^{(t)} - \alpha \frac{dE}{dw_{ij}^{(t)}}$$

Exploding Gradients

If the weights in a network are very large, then the gradients for the lower layers involve products of many large terms. In this case you can have **exploding** gradients: gradients that get too large to converge.

Solution: Lowering the learning rate can help prevent exploding gradients.

Detecting Overfitting for Neural Networks

A learning curve (or training curve) plots a model's loss against the number of training iterations (epochs). Notice that the two loss curves below behave similarly at first and then diverge. That is, after a certain number of iterations, loss declines or holds steady (converges) for the training set, but increases for the validation set. This suggests **overfitting**.

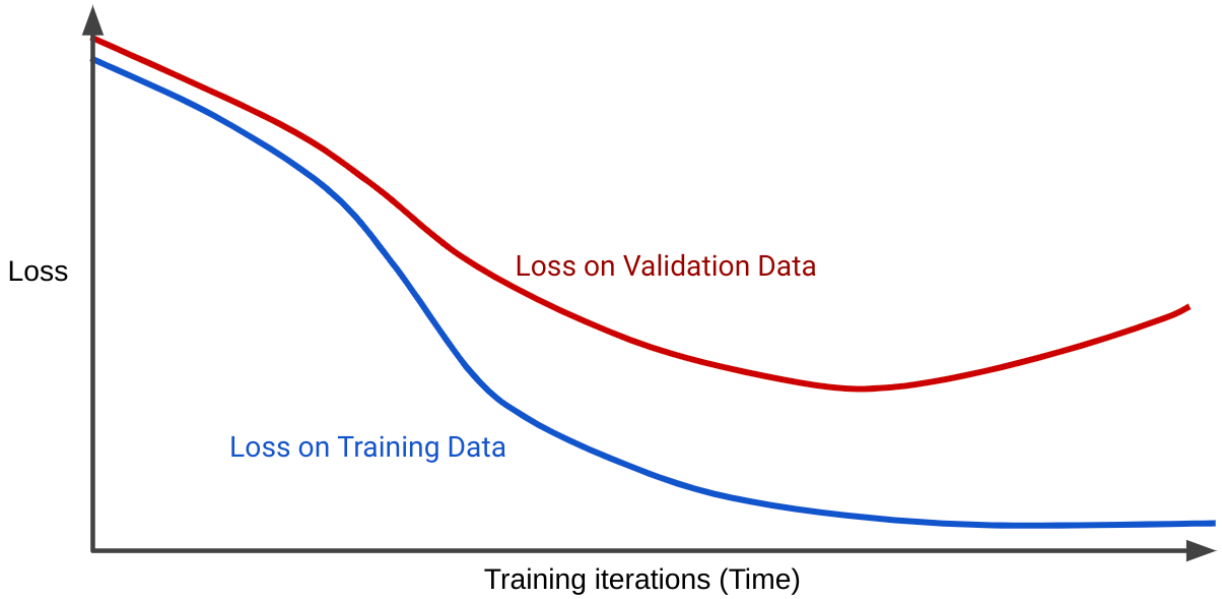


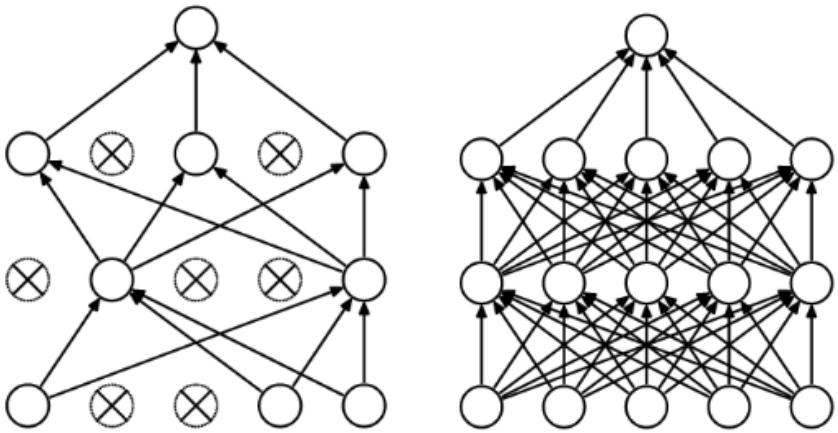
Image credit: Google

Reasons for overfitting:

- Model is too complex
- Training for too many iterations
- Training set is too small
- Presence of noise in data

Popular Methods to Overcome Overfitting for Neural Nets

Dropout layers



During training
After applying dropout

For final prediction
and inference

- Neurons are randomly dropped during training to reduce overfitting.
- At inference, all neurons are used with scaled outputs.
- The **dropout rate** sets the probability of dropping a neuron.

Early stopping

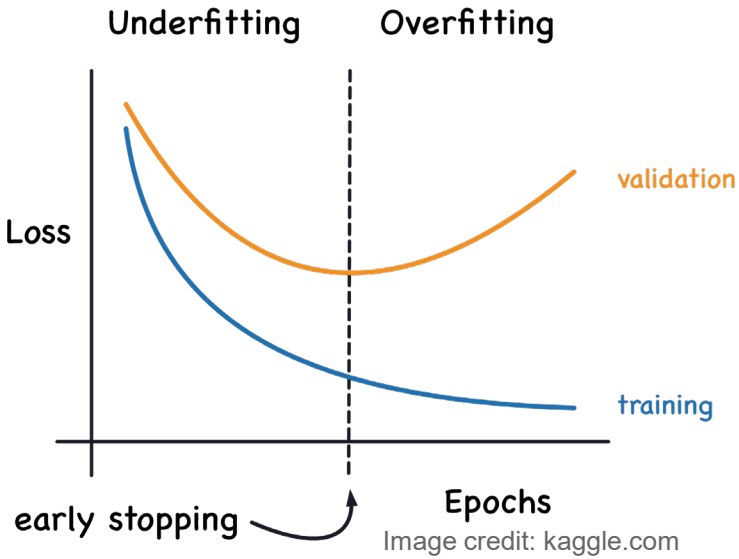
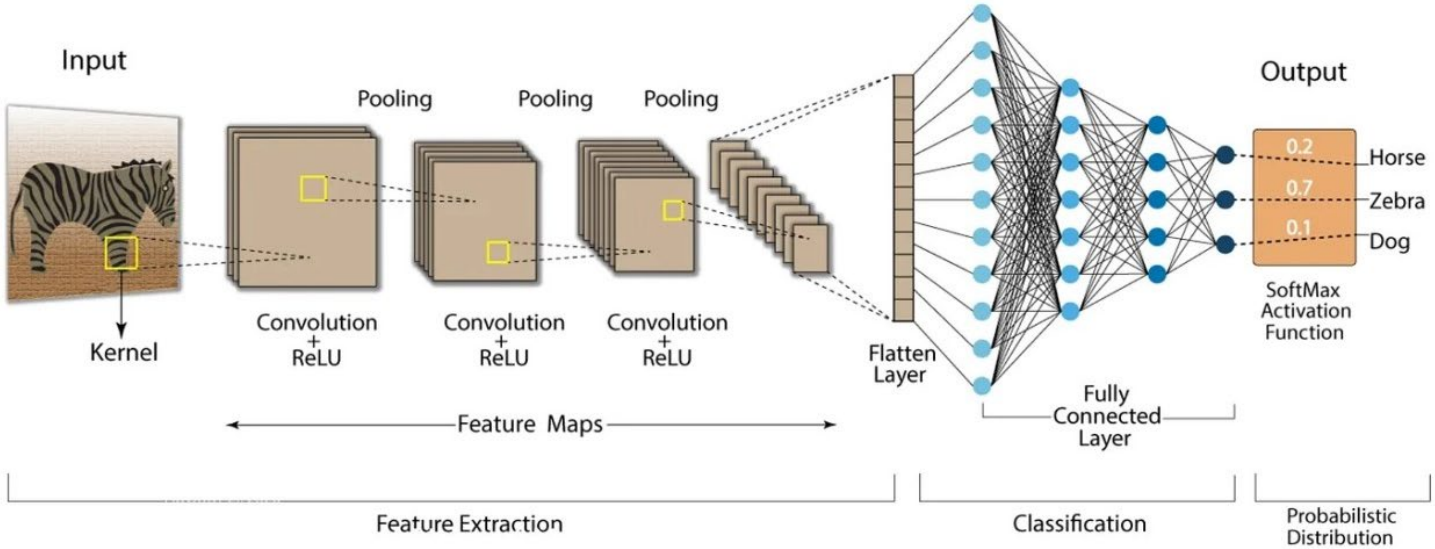
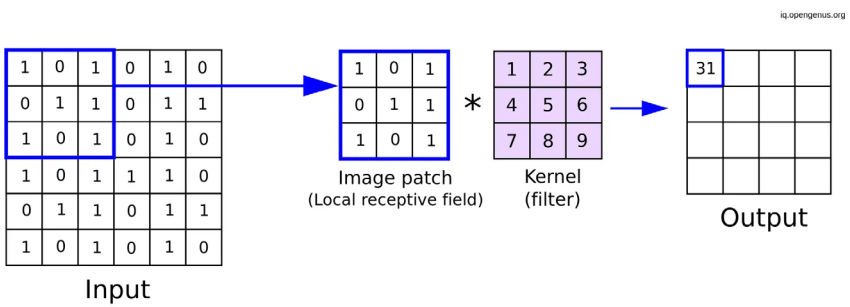


Image credit: kaggle.com

- Early stopping halts training when validation performance stops improving, preventing overfitting.
- Main parameter is **patience** – how many epochs to wait after no improvement.

Advanced Architecture: Convolutional Neural Network



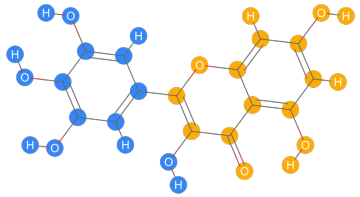


- Excellent for grid-like data (images, spectra, 2D/3D fields).
- Captures spatial/local correlations via filters via convolutional operations.
- Pooling layers reduce dimensionality while retaining important features.

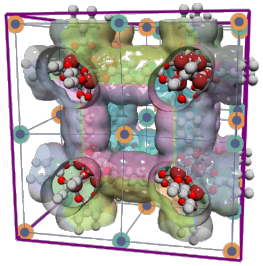
- **Applications in chemical engineering**: Analyzing microscopy images (crystal morphology, catalyst particles). Spectroscopy data interpretation (Raman, IR). Predicting material properties from structural images / 3D atomic fields.

Image credit: WWW

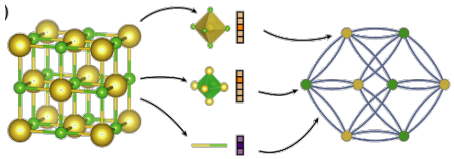
Advanced Architecture: Graph Convolutional Neural Network



Molecular graph
(image credit: Wolfram)

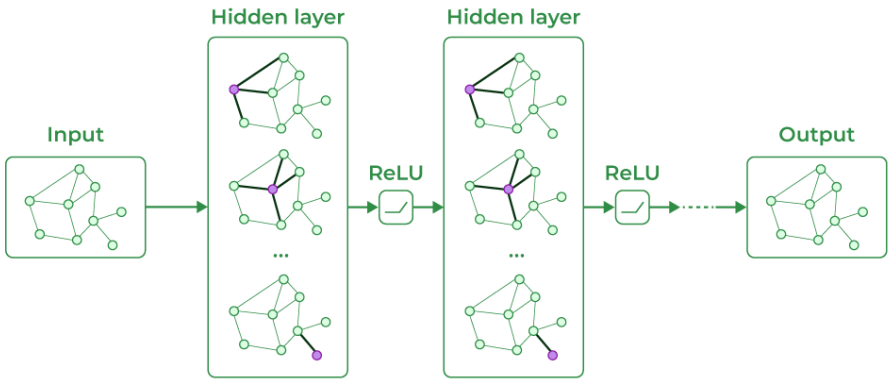


Pore graph
(developed by the Shi group)



Crystal graph
(Xie et al., Phys. Rev. Lett. 120(14), 145301 (2018))

Knowledge graph, social graph and many more…



Graph convolution
(image credit: GeeksforGeeks)

- Generalize the convolution operation of CNNs to graph-structured data (molecules, materials, pore networks).
- Captures relational/topological information.
- **Applications in chemical engineering**: Molecular property prediction (drug discovery, catalysts).Crystal/porous material design (MOFs, zeolites). Reaction network modeling. Knowledge graph enhanced LLM
- A Gentle Introduction to Graph Neural Networks

Advanced Architecture: Recurrent Neural Network

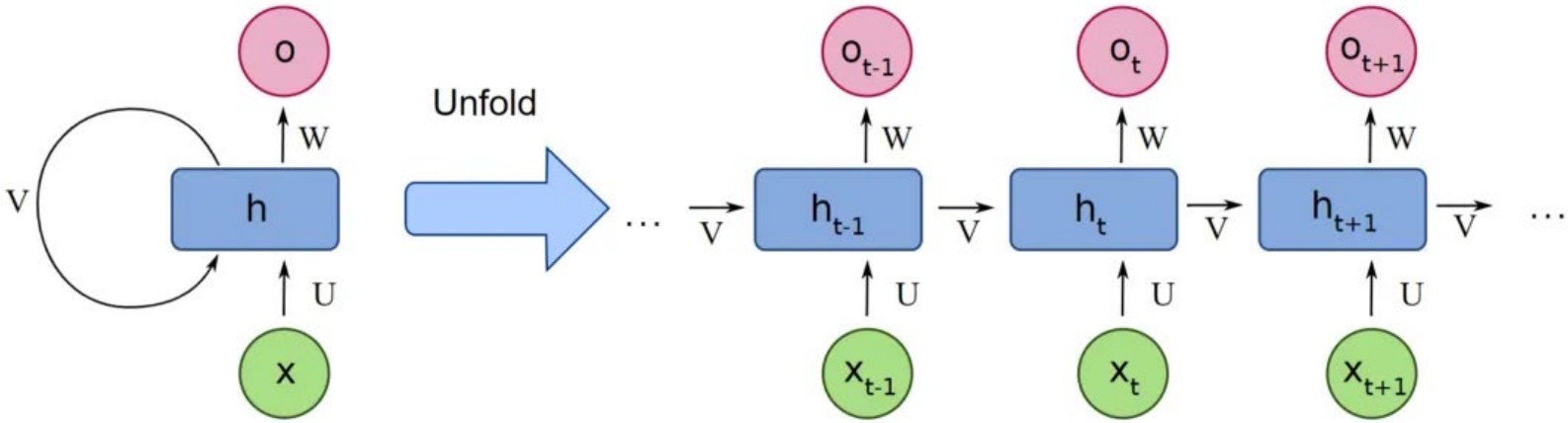


Image credit: Analytics Vidhya

- Designed for sequential/time-series data.
- Good when order and temporal dependence matter.
- Maintains hidden states that carry information from previous steps; can model dependencies across time.
- **Applications in chemical engineering**: Process monitoring and fault detection (plant sensor data). Predicting molecular sequences (e.g., polymer chains).

Advanced Architecture: Residual Neural Network (ResNet)

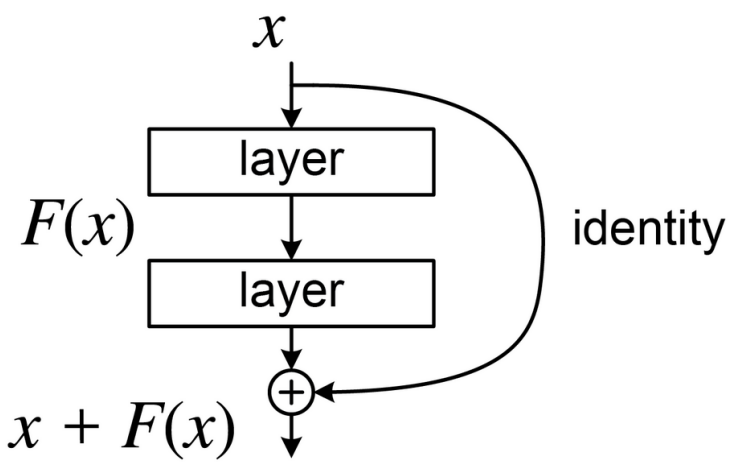


Image credit: Wikipedia

- Skip (residual) connections let very deep networks train without vanishing gradients.
- The skip/shortcut connection passes the input x forward unchanged, so the layer only has to learn what is missing (the residual).
- **Applications**: High-resolution image/spectral analysis, microstructure and tomography in materials. Basically, ResNet shines when you need very deep feature extraction from images or spectral data.

Hands-on Module:

Prediction of Hydrogen Storage Capacity in Metal-Organic Frameworks

University at Buffalo
The State University of New York

Urgent needs for clean energy storage technologies towards a net zero 2050

Green hydrogen production and application

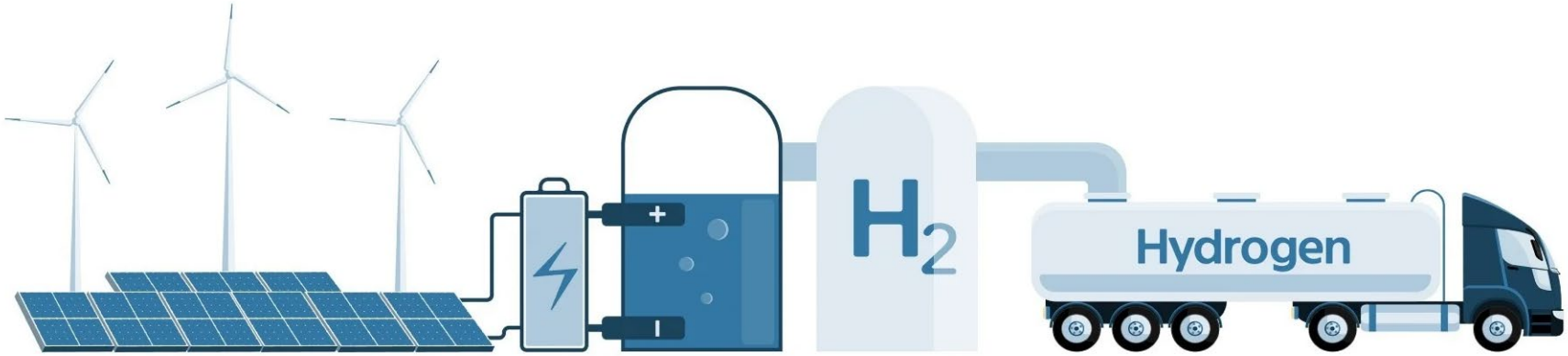
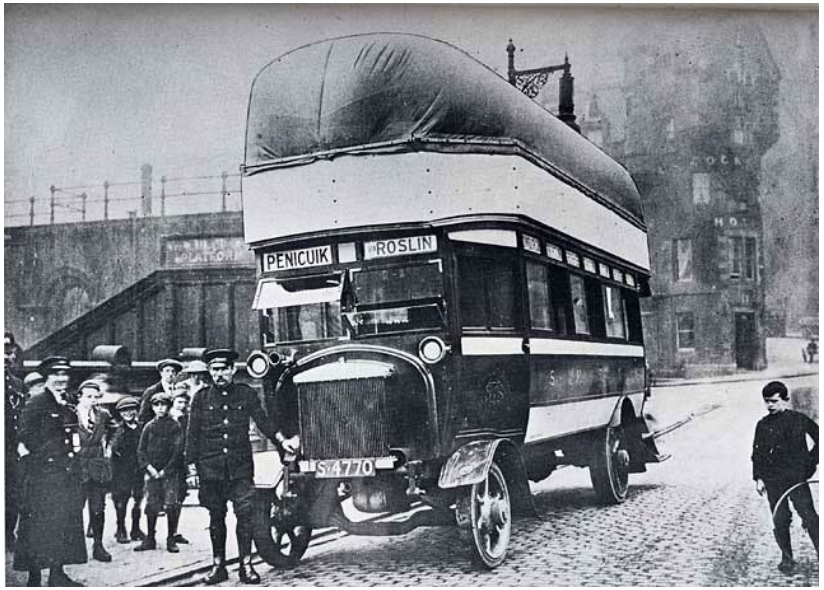


Image Credit: Scharfsinn/Shutterstock.com

- **Green hydrogen**: Produced by water electrolysis using renewable energy (wind, solar, hydro); zero carbon emissions.
- **Blue hydrogen**: Derived from natural gas with carbon capture and storage (CCS) to reduce CO_2 emissions.
- **Gray hydrogen**: Produced from fossil fuels (mainly natural gas via steam methane reforming) without CCS.
- **Brown hydrogen**: Generated from coal gasification; most carbon-intensive method.

Historical Gas Storage for Vehicles

A gas bag vehicle from World War I.



Gas bag bus operated by the Scottish Motor Traction Company, 1914-1918.

The old-timers on these pictures are not moving furniture or an oversized load. What can be seen on the roof is the fuel tank of the vehicle - a balloon filled with uncompressed gas.

https://solar.lowtechmagazine.com/2011/11/gas-bag-vehicles

Historical Gas Storage for Vehicles

This Dutch old-timer carries a gas storage bag of 13 m^3, which gave it a range of approximately 50 km (30 miles)

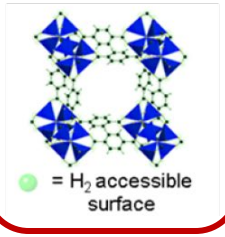
https://solar.lowtechmagazine.com/2011/11/gas-bag-vehicles

Current Hydrogen Storage Technology

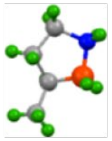
How is hydrogen stored?

- **Physical-based**
  - Compressed Gas
  - Cold/Cryo Compressed
  - Liquid H_2

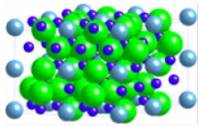
- **Material-based**
  - Adsorbent
    - Ex. MOF-5



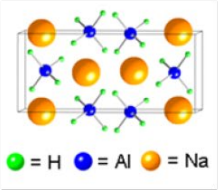
  - Liquid organic
    - Ex. BN-methyl cyclopentane



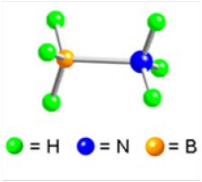
  - Interstitial hydride
    - Ex. $LaNi_5H_6$



  - Complex hydride
    - Ex. $NaAlH_4$



  - Chemical hydrogen
    - Ex. NH_3BH_3



Toyota Mirai

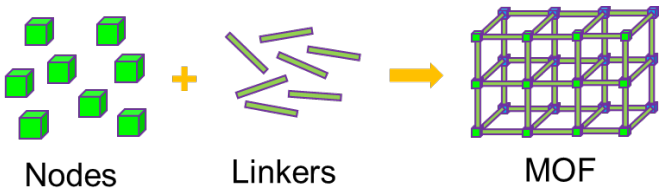
High-pressure hydrogen tank in Mirai (700 bar)

https://www.energy.gov/eere/fuelcells/hydrogen-storage

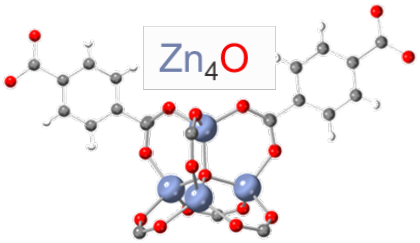
Metal-Organic Frameworks (MOFs)

Metal-organic frameworks (MOFs) are composed of inorganic "nodes" connected by organic "linkers" to form a periodic framework material.

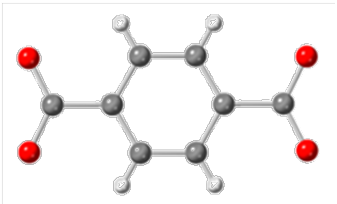
- Highly porous and crystalline
- Highly tunable structures and chemistry



Inorganic node:

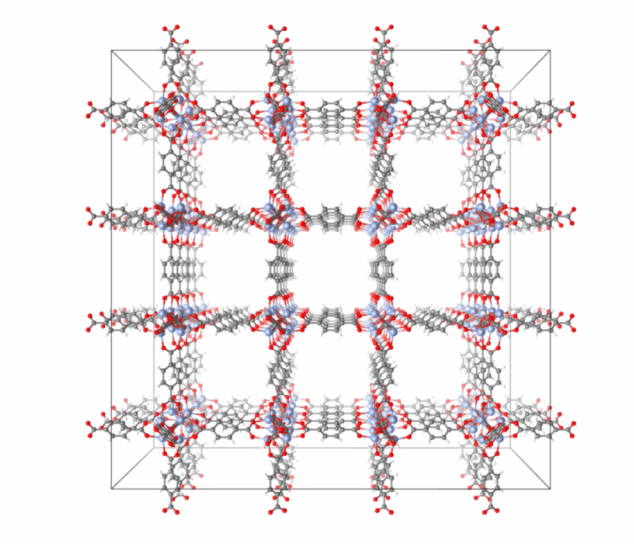


Organic linkers:



bdc^{2-}

bdc = 1,4-benzendicarboxylate



MOF-5 (IRMOF-1)

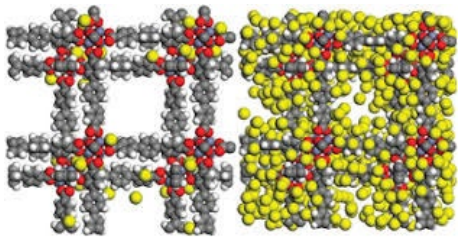
$Zn_4O(bdc)_3$

slides courtesy of Andrew Rosen

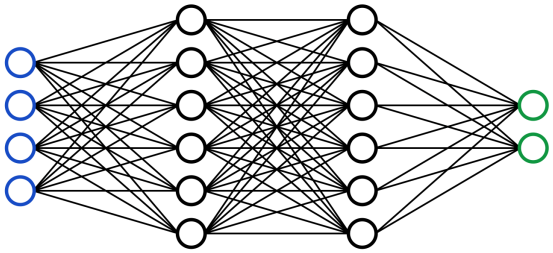
Fast Evaluation of a Large Number of MOFs

- Over **100,000** experientially synthesized MOFs
- Over **1,000,000** computationally assembled (hypothetical) MOFs

Adsorption experiments
(**days to weeks** per MOF)



Molecular simulation
(**hours to days** per MOF)



ML model
(**<<1 second** per MOF)

Image credit: WWW

Hands-on: Building a Multilayer Perceptron for Prediction of Hydrogen Storage in MOFs

See Google Colab Notebook